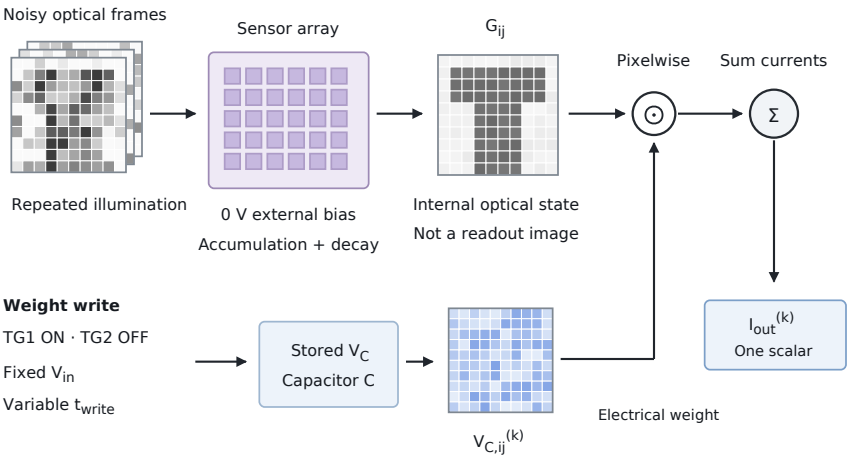
Zero-bias photosensing and voltage-weighted computing

GaON/GaN photomemristive sensor · 2TG1C1S pixel array

a Optical state × electrical weight → scalar output



Readout: TG1 OFF · TG2 ON

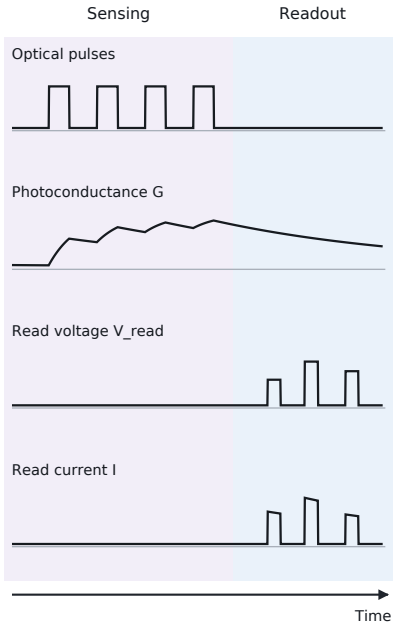
$I_{ij}(t_r) \approx G_{ij}(t_r) \cdot V_{C,ij}(t_r)$

$I_{out}^{(k)} \approx \Sigma_{ij}\, G_{ij} \cdot V_{C,ij}^{(k)}$

Template k → one scalar; change template → next output

Short, approximately ohmic readout; small capacitor droop.

b Device response



Schematic traces; no measured values

Concept schematic · optical state and capacitor voltage are distinct · author scientific review required